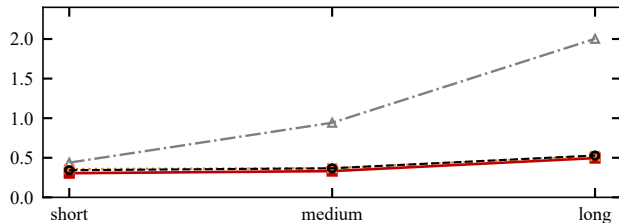


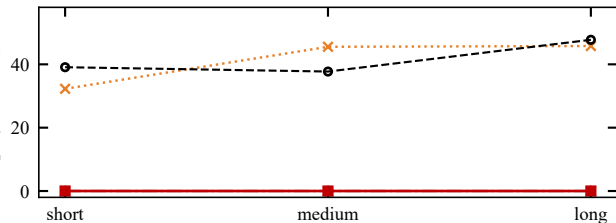
▲ - clean solve ■ - AgentTX causal (ours) ✕ - optimistic checkpoint ● - whole-branch abort

Median charged tokens (millions)



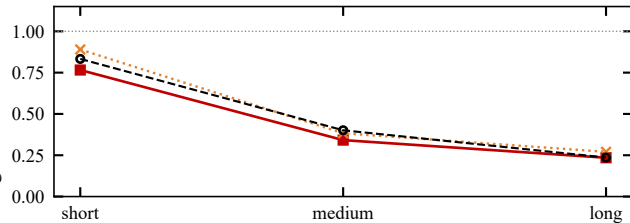
SWE tasks by clean-solve cost
(a) Session + replay

Median replay tokens (thousands)



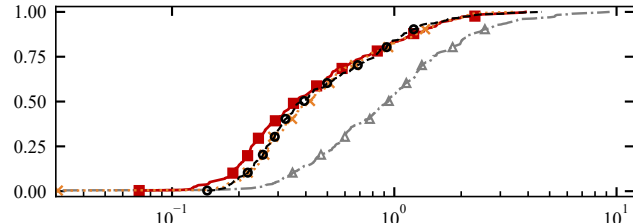
SWE tasks by clean-solve cost
(b) Replay tax

Charged tokens / clean solve



SWE tasks by clean-solve cost
(c) Paired ratio

CDF



Charged tokens (millions)
(d) Charged-token CDF